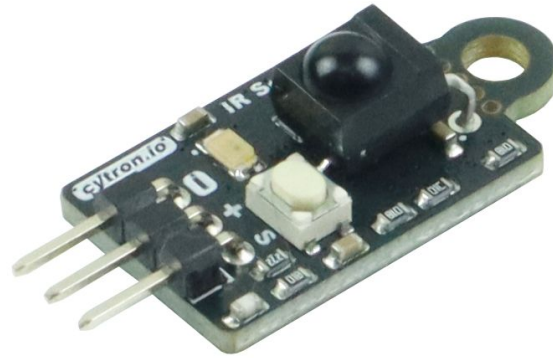
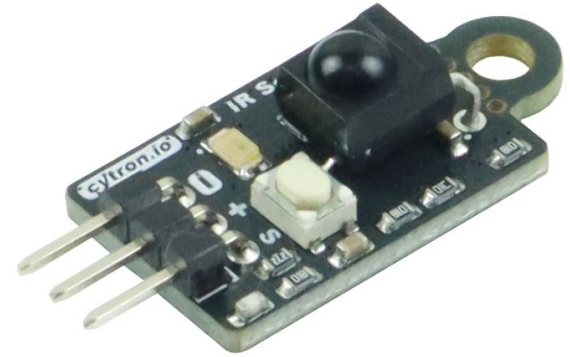




Pengenalan

Apa itu IR Sumo Start?

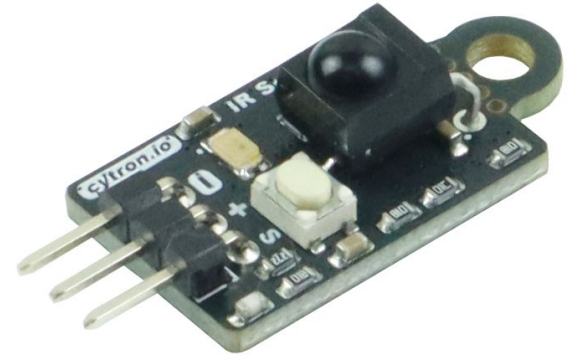
- IR Sumo Start adalah satu modul yang digunakan untuk memberi signal START/STOP kepada robot sumo auto melalui remote TV (signal IR).
- Ia digunakan di pertandingan sumo peringkat antarabangsa di Jepun.



Pengenalan

Kelebihan menggunakan IR Sumo Start

- 2 robot yang berlawanan bergerak serentak. Tiada isu robot bergerak lebih awal.
- Keselamatan lebih terjamin. Robot boleh dihentikan secara wireless.



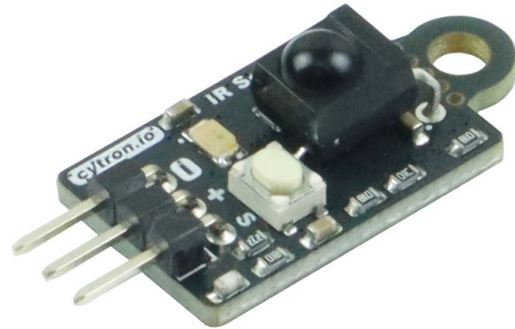
Datasheet



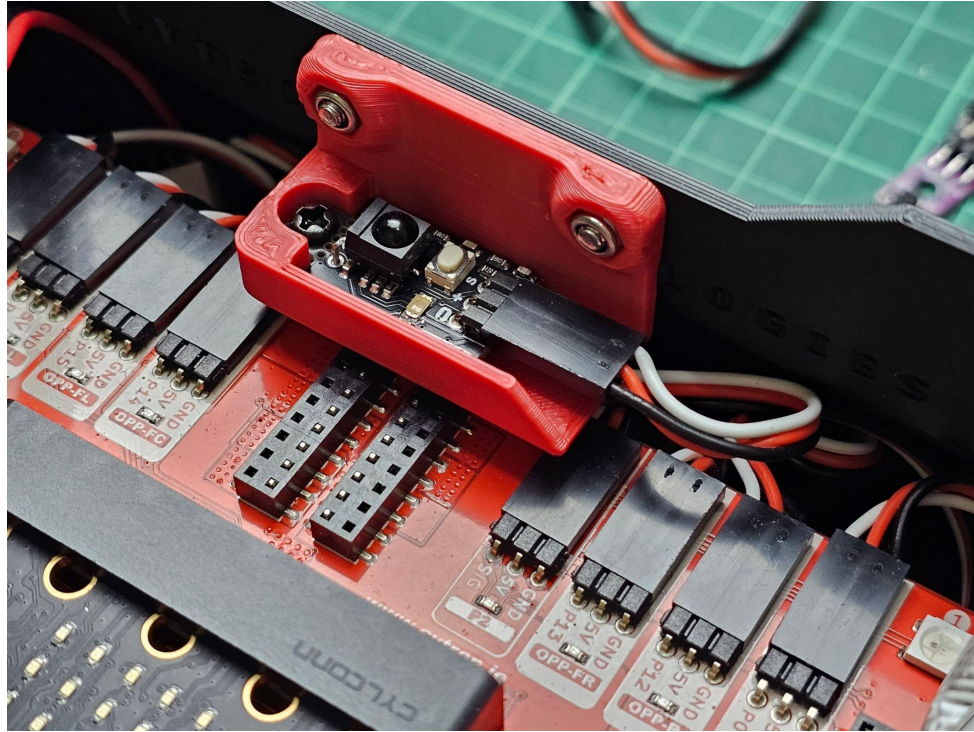
Pemasangan

Pastikan IR Sumo Start dipasang dalam keadaan

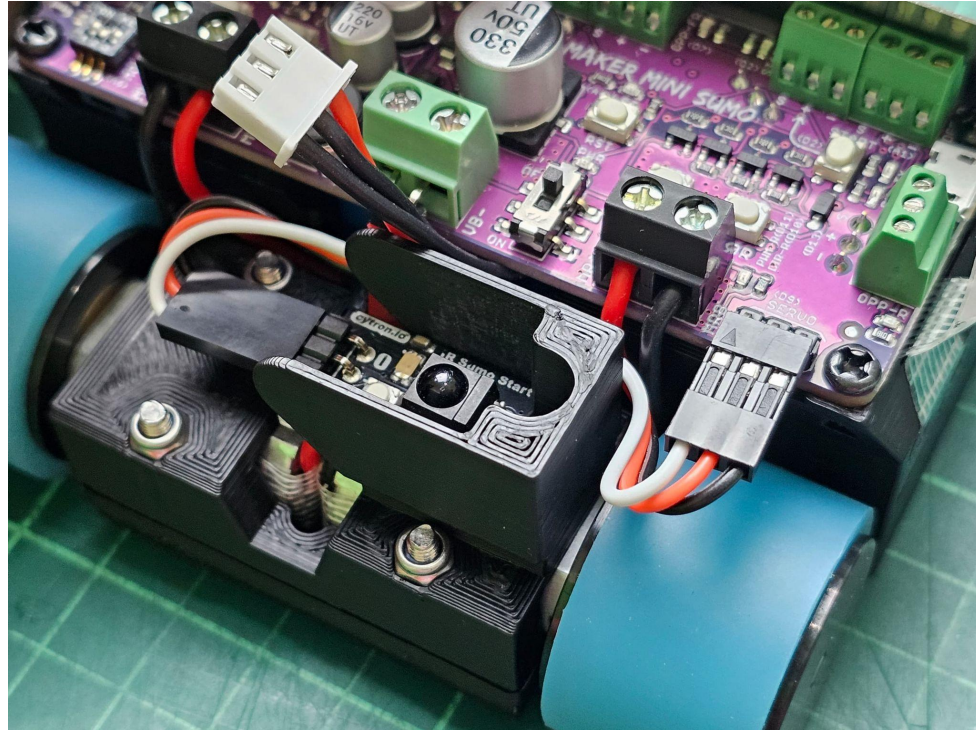
- terhindar daripada noise.
- mudah di akses.
- sudut paling optimum untuk terima signal daripada juri.



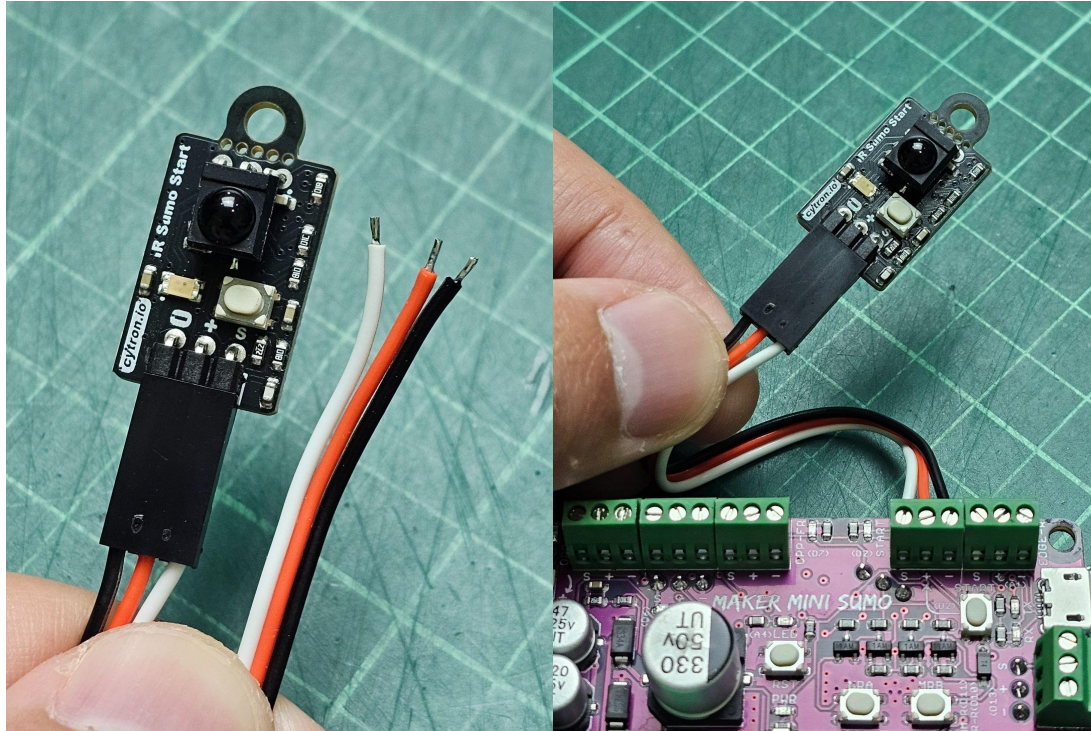
Pemasangan



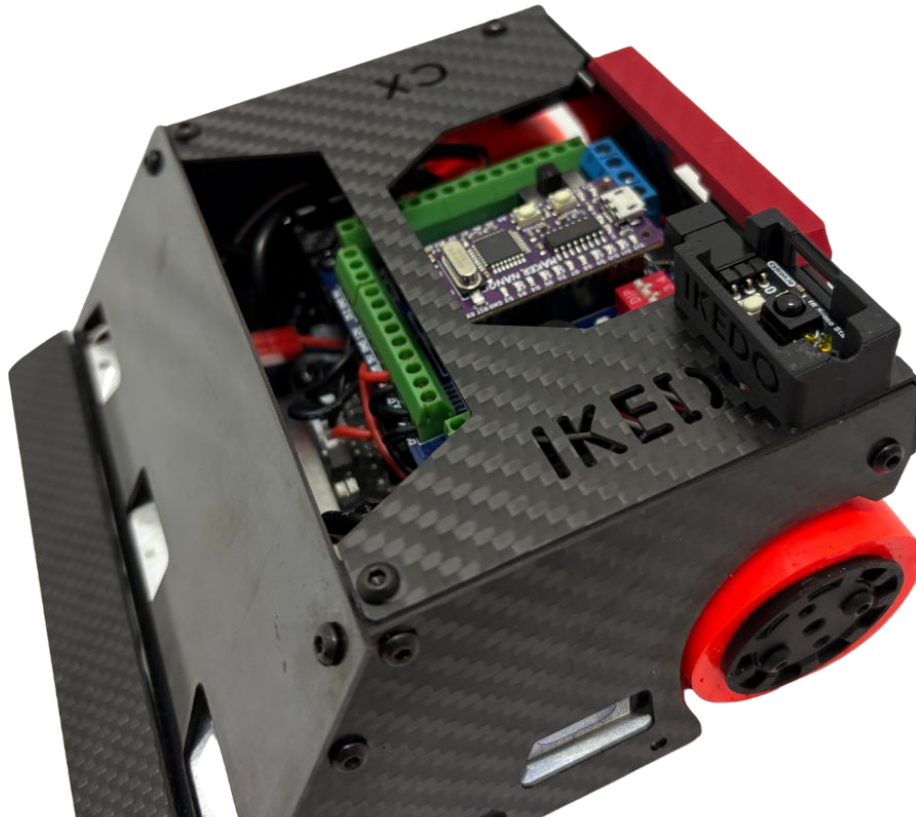
Pemasangan



Pemasangan



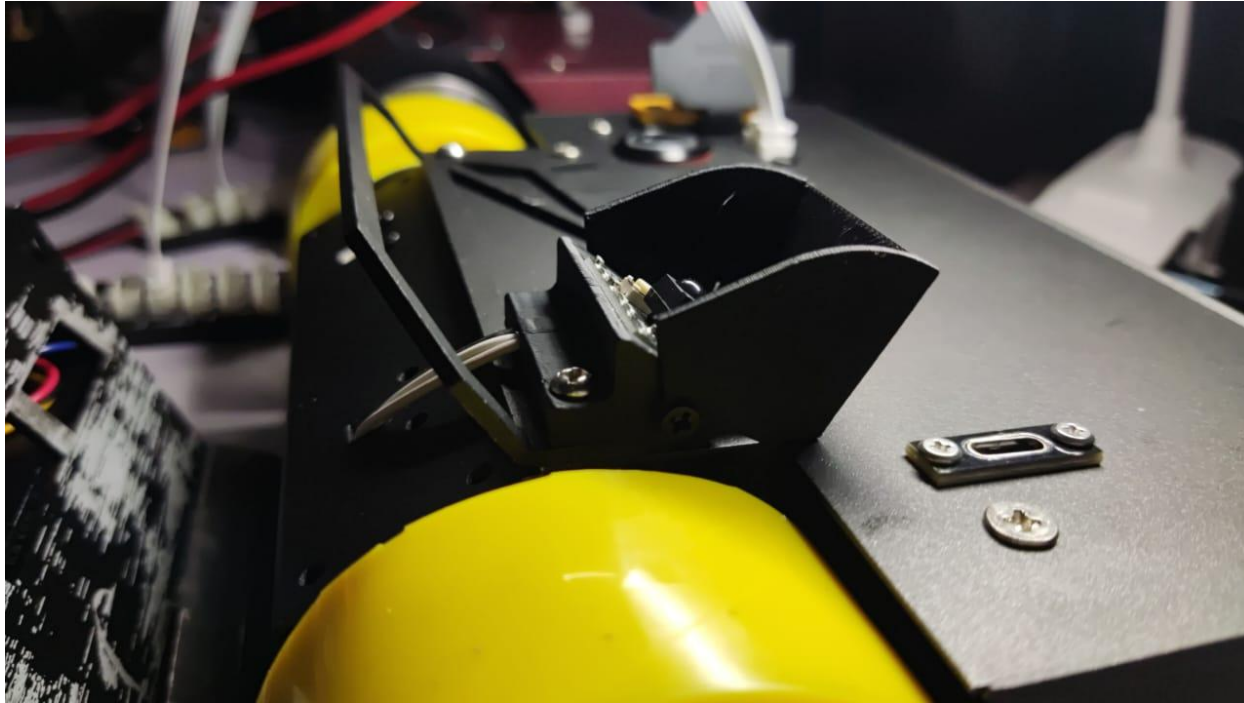
Pemasangan



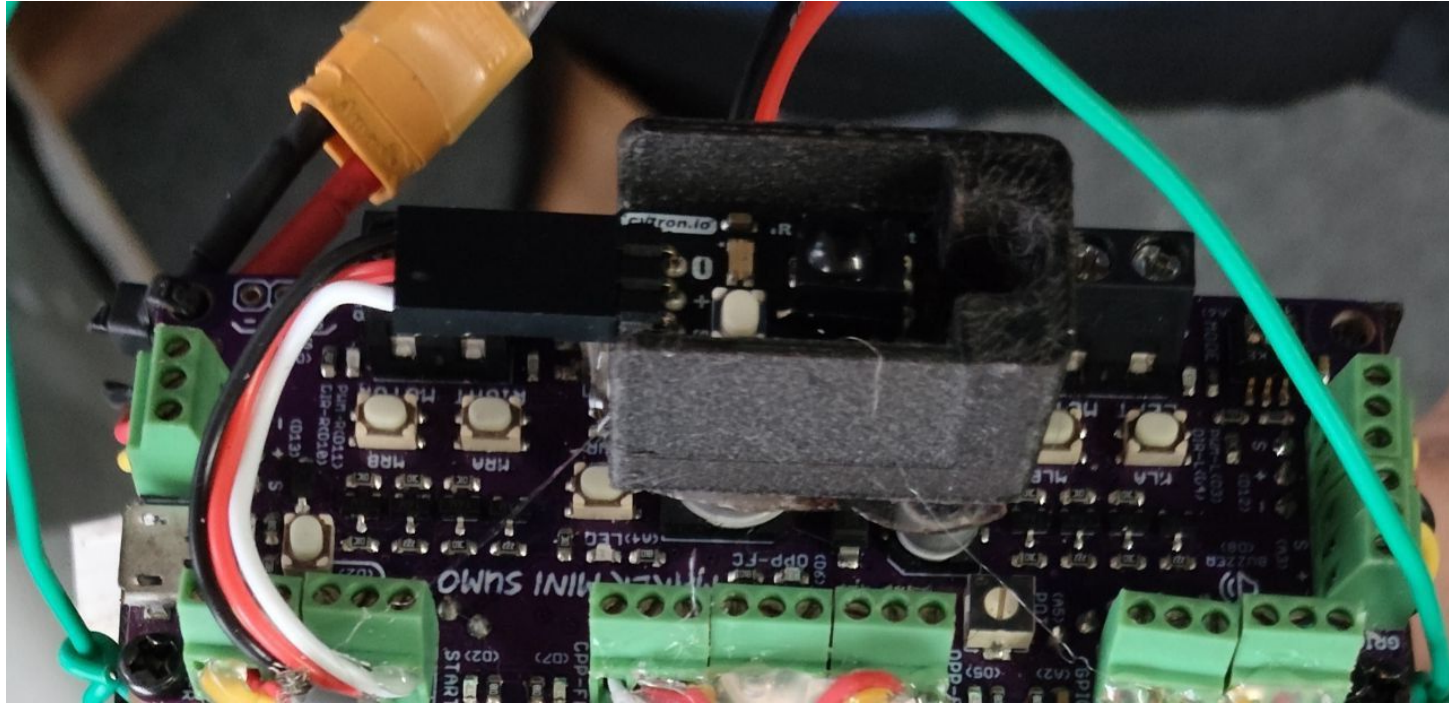
Pemasangan



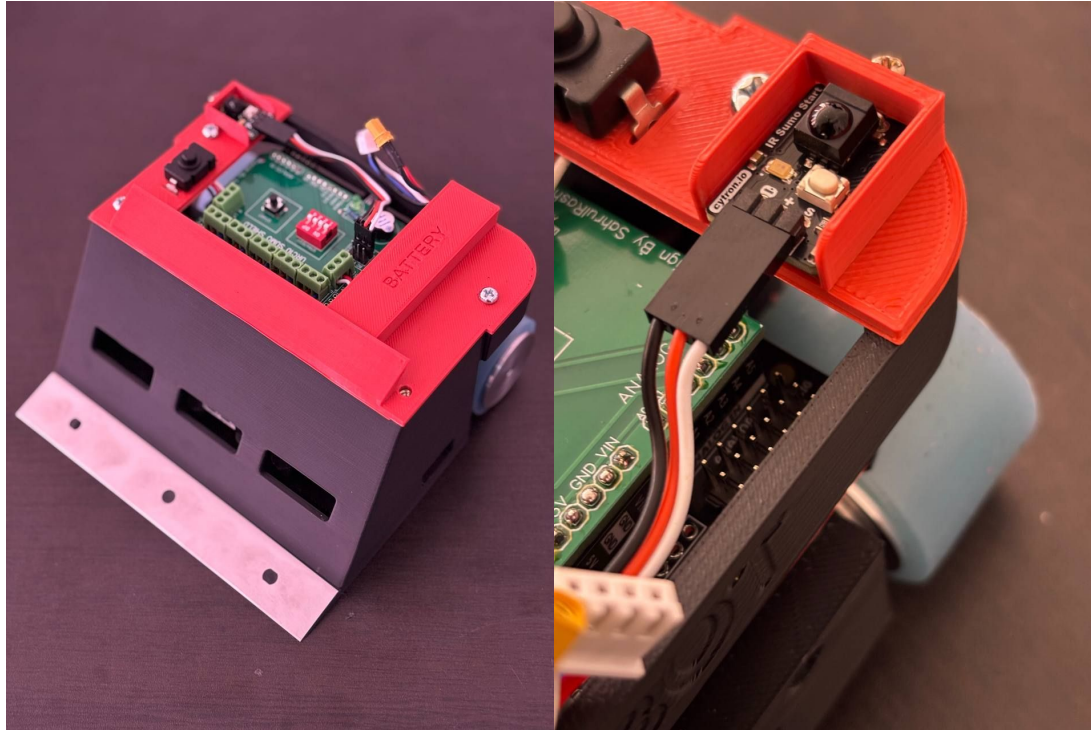
Pemasangan



Pemasangan



Pemasangan



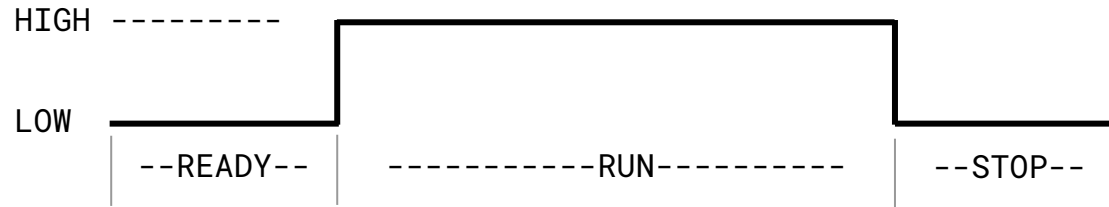
Membina Kod Pengaturcaraan

- Signal daripada IR Sumo Start adalah digital (LOW/HIGH).

Remote	State	LED indicator	Signal
	Power up	LED hijau menyala 1 saat dan padam.	LOW
Key 1	READY	LED merah terang (pink) flashing cepat, kemudian LED biru menyala statik.	LOW
Key 2	RUN	LED hijau menyala statik.	HIGH
Key 3	STOP	LED merah blinking perlahan.	LOW
	Noise	LED biru terang flashing	Tiada perubahan

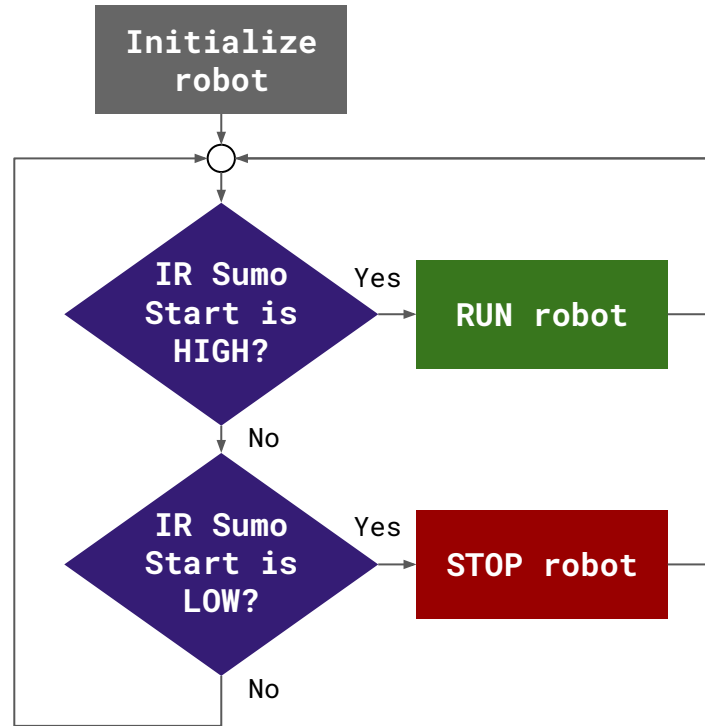
Membina Kod Pengaturcaraan

- Graf ringkas berkenaan signal daripada IR Sumo Start.



Membina Kod Pengaturcaraan

1.



Membina Kod Pengaturcaraan

```
#define IR_SUMO_START 9

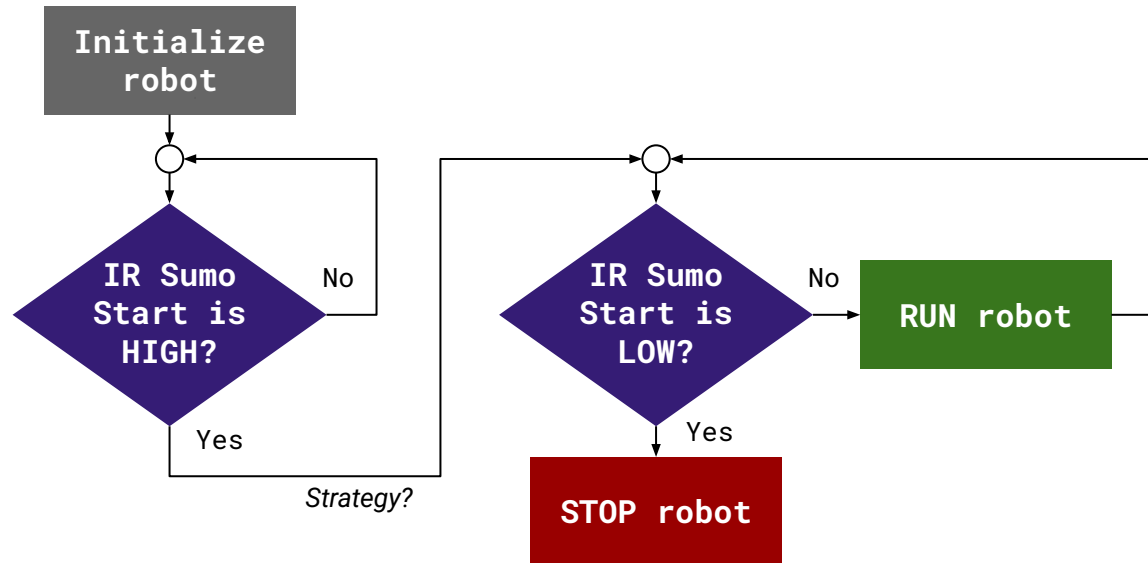
void setup()
{
  MakerSumo.begin(); // Initialize robot

  pinMode(IR_SUMO_START, INPUT);
}

void loop()
{
  if (digitalRead(IR_SUMO_START) == HIGH)
  {
    // Run the robot
    strategy();
    search();
    attack();
  }
  else
  {
    // Stop the robot
    MakerSumo.stop();
  }
}
```

Membina Kod Pengaturcaraan

2.



Membina Kod Pengaturcaraan

```
#define IR_SUMO_START 9

void setup()
{
  MakerSumo.begin(); // Initialize robot

  pinMode(IR_SUMO_START, INPUT);

  // Tunggu sehingga terima signal RUN
  while (!digitalRead(IR_SUMO_START));

  // Run the robot
  strategy();
}
```

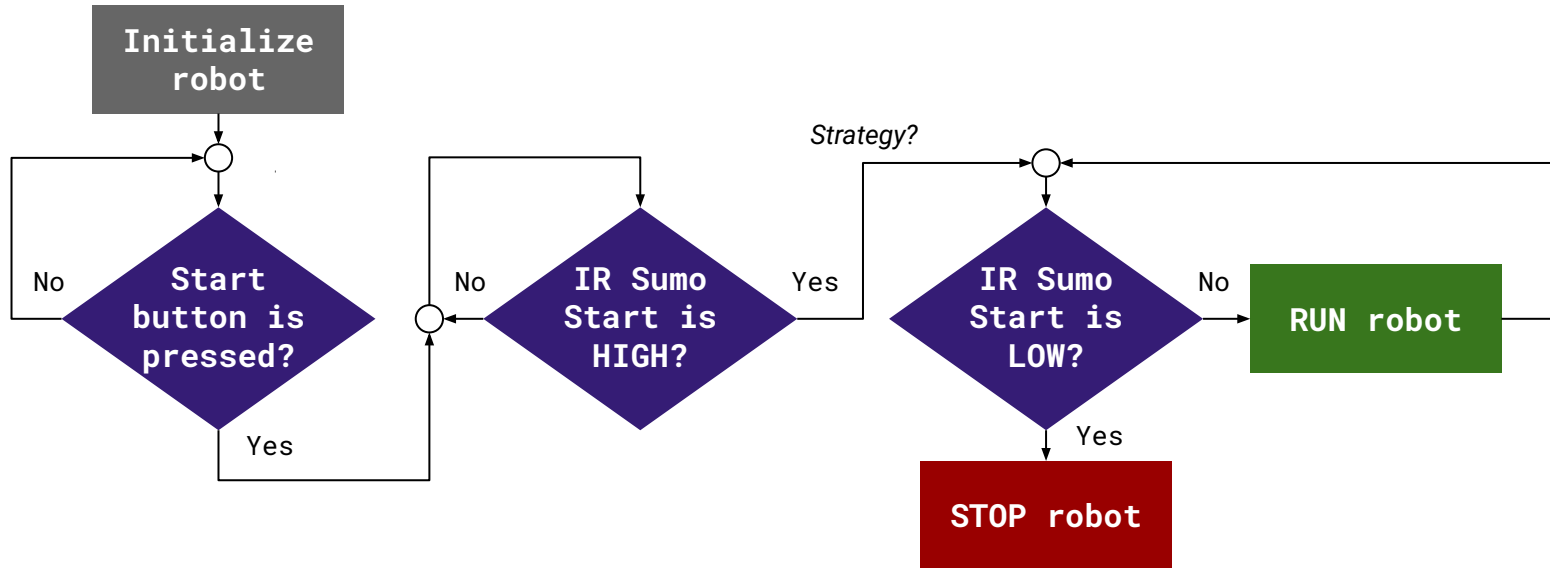
```
void loop()
{
  // Run the robot
  search();
  attack();

  if (digitalRead(IR_SUMO_START) == LOW)
  {
    // Stop the robot
    MakerSumo.stop();

    while (true);
  }
}
```

Membina Kod Pengaturcaraan

3.



Membina Kod Pengaturcaraan

```
#define IR_SUMO_START 9

void setup()
{
  MakerSumo.begin(); // Initialize robot

  pinMode(IR_SUMO_START, INPUT);

  // Tunggu sehingga button START ditekan
  while (digitalRead(START));

  // Tunggu sehingga terima signal RUN
  while (!digitalRead(IR_SUMO_START));

  // Run the robot
  strategy();
}

void loop()
{
  // Run the robot
  search();
  attack();

  if (digitalRead(IR_SUMO_START) == LOW)
  {
    // Stop the robot
    MakerSumo.stop();

    while (true);
  }
}
```

Video & Sample Code based on Ikedo Mini V2

